CampusLaundry Management System

Data Definition Language (DDL) Specification & Relational Schema

Project: CampusLaundry Management System	**Category:** DDL Commands & Schema Architecture	**DBMS Engine:** MySQL 8.0+ / InnoDB
Schema Normalization: 3NF (Strict Third Normal Form)	**Academic Scope:** Database Management Systems Laboratory	**Deployment:** Cloud Production Verified (Aiven)

1. Schema Architecture & 3NF Normalization Principles

The CampusLaundry database schema is architected to satisfy **Boyce-Codd / Third Normal Form (3NF)**. A critical architectural refinement was the removal of Vendor_ID from the laundry_transaction table. Because every drop point kiosk is uniquely mapped to a servicing vendor via drop_point.Vendor_ID, storing Vendor_ID directly in laundry_transaction introduced a transitive functional dependency: Transaction_ID → Drop_Point_ID → Vendor_ID. Removing this redundancy eliminates update and deletion anomalies.

2. Database Initialization DDL

Initializes the database catalog with UTF-8 Unicode 4-byte character support (utf8mb4) and case-insensitive collation.

```
-- Database Creation and Context Selection
CREATE DATABASE IF NOT EXISTS laundry
    CHARACTER SET utf8mb4
    COLLATE utf8mb4_0900_ai_ci;

USE laundry;
```

3. Entity Table Definitions & Integrity Constraints

3.1 Table: students (Student Quota & Profile Ledger)

Tracks student demographic profiles and maintains semester laundry cycle allowances (15 or 30 cycles per 8-month academic year).

```
CREATE TABLE IF NOT EXISTS students (
    std_id VARCHAR(10) PRIMARY KEY,
    std_name VARCHAR(60) NOT NULL,
    contact BIGINT NOT NULL,
    hostel VARCHAR(50) NOT NULL,
    room VARCHAR(20) NOT NULL,
    academic_year VARCHAR(30) NOT NULL,
    quota INT NOT NULL DEFAULT 30,
    cycle_used INT NOT NULL DEFAULT 0,
    cycle_remaining INT NOT NULL DEFAULT 30,
    CONSTRAINT chk_quota_val CHECK (quota IN (15, 30)),
    CONSTRAINT chk_cycles_balance CHECK (cycle_used + cycle_remaining = quota)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

3.2 Table: vendor (Third-Party Laundry Contractors)

Maintains master vendor contracting data. Vendors manage commercial washing, ironing, and delivery for assigned kiosks.

```
CREATE TABLE IF NOT EXISTS vendor (
    Vendor_ID VARCHAR(20) PRIMARY KEY,
    Vendor_Name VARCHAR(50) NOT NULL,
    Contact_No BIGINT NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

3.3 Table: drop_point (Campus Kiosks & Lockers)

Represents physical drop-off and collection kiosks placed at hostel clusters. Foreign key links each kiosk to its assigned servicing vendor.

```
CREATE TABLE IF NOT EXISTS drop_point (
    Drop_Point_ID VARCHAR(20) PRIMARY KEY,
    Location VARCHAR(50) NOT NULL,
    Operational_Status VARCHAR(20) NOT NULL DEFAULT 'Active',
    Vendor_ID VARCHAR(20) NOT NULL,
    CONSTRAINT fk_vendor_id FOREIGN KEY (Vendor_ID)
        REFERENCES vendor(Vendor_ID) ON DELETE CASCADE ON UPDATE CASCADE,
    CONSTRAINT chk_dp_status CHECK (Operational_Status IN ('Active', 'Under Maintenance', 'Inactive'))
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

3.4 Table: laundry_item (Garment Catalog & Benchmark Weights)

Catalog of standard clothing categories with benchmark weight coefficients used to calculate batch laundry loads (e.g., 6.0 kg max per cycle).

```
CREATE TABLE IF NOT EXISTS laundry_item (
    Item_ID VARCHAR(20) PRIMARY KEY,
    Item_Name VARCHAR(50) NOT NULL,
    Category VARCHAR(30) NOT NULL,
    Standard_Weight DECIMAL(5,2) NOT NULL DEFAULT 0.25,
    CONSTRAINT chk_item_weight CHECK (Standard_Weight > 0.00)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

3.5 Table: laundry_transaction (Service Lifecycle Ledger — 3NF)

Records individual student laundry requests. Notice: Vendor_ID is omitted in compliance with 3NF. The servicing vendor is determined via drop_point.Vendor_ID.

```
CREATE TABLE IF NOT EXISTS laundry_transaction (
    Transaction_ID VARCHAR(20) PRIMARY KEY,
    std_id VARCHAR(10) NOT NULL,
    Drop_Point_ID VARCHAR(20) NOT NULL,
    Drop_Off_Date DATE NOT NULL,
    Expected_Completion_Date DATE NOT NULL,
    Total_Weight DECIMAL(5,2) NOT NULL DEFAULT 0.00,
    Cycles_Deducted INT NOT NULL DEFAULT 1,
    Status VARCHAR(20) NOT NULL DEFAULT 'Submitted',
    Actual_Completion_Date DATE NULL,
    CONSTRAINT fk_std_id FOREIGN KEY (std_id)
        REFERENCES students(std_id) ON DELETE CASCADE ON UPDATE CASCADE,
    CONSTRAINT fk_drop_point_id FOREIGN KEY (Drop_Point_ID)
        REFERENCES drop_point(Drop_Point_ID) ON DELETE RESTRICT ON UPDATE CASCADE,
    CONSTRAINT chk_txn_status CHECK (
        Status IN ('Submitted', 'In Washing', 'Ready for Pickup', 'Completed', 'Cancelled')
    )
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

3.6 Table: transaction_item (Composite Junction Entity)

Decomposes the many-to-many relationship between laundry_transaction and laundry_item. Enforces a composite primary key (Transaction_ID, Item_ID) with cascading delete.

```
CREATE TABLE IF NOT EXISTS transaction_item (
    Transaction_ID VARCHAR(20) NOT NULL,
    Item_ID VARCHAR(20) NOT NULL,
    Quantity INT NOT NULL DEFAULT 1,
    PRIMARY KEY (Transaction_ID, Item_ID),
    CONSTRAINT fk_transaction_id FOREIGN KEY (Transaction_ID)
        REFERENCES laundry_transaction(Transaction_ID) ON DELETE CASCADE ON UPDATE CASCADE,
    CONSTRAINT fk_item_id FOREIGN KEY (Item_ID)
        REFERENCES laundry_item(Item_ID) ON DELETE RESTRICT ON UPDATE CASCADE,
    CONSTRAINT chk_item_quantity CHECK (Quantity > 0)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

3.7 Table: admin_users (Role-Based Access Control Ledger)

Stores credentials for wardens, vendor operators, and administrative staff. Passwords stored as 60-byte bcrypt hashes.

```
CREATE TABLE IF NOT EXISTS admin_users (
    admin_id INT AUTO_INCREMENT PRIMARY KEY,
    username VARCHAR(30) NOT NULL UNIQUE,
    password_hash VARCHAR(255) NOT NULL,
    full_name VARCHAR(60) NOT NULL,
    role ENUM('warden', 'vendor', 'superadmin') NOT NULL DEFAULT 'warden',
    is_active BOOLEAN NOT NULL DEFAULT TRUE,
    last_login DATETIME NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

4. Schema Alteration & Evolution DDL

Production migration statements used to enforce 3NF normalization and enhance query indexing across transaction tables.

```
-- 1. Normalization Refactoring: Remove Transitive Vendor Dependency
ALTER TABLE laundry_transaction DROP FOREIGN KEY fk2_vendor_id;
ALTER TABLE laundry_transaction DROP COLUMN Vendor_ID;

-- 2. Query Performance Optimization: Secondary Composite Indexes
CREATE INDEX idx_txn_student_status ON laundry_transaction(std_id, Status);
CREATE INDEX idx_txn_dropoff_date ON laundry_transaction(Drop_Off_Date DESC);
CREATE INDEX idx_drop_point_vendor ON drop_point(Vendor_ID);
CREATE INDEX idx_admin_user_active ON admin_users(username, is_active);
```

5. Modular & Composite View DDL Definitions

High-performance database views encapsulate complex queries and power student and executive dashboards without redundant code.

5.1 View: vw_student_profile

```
CREATE OR REPLACE VIEW vw_student_profile AS
SELECT
    std_id,
    std_name,
    contact,
    hostel,
    room,
    academic_year
FROM students;
```

5.2 View: vw_student_quota_burn (Burn Analytics)

```
CREATE OR REPLACE VIEW vw_student_quota_burn AS
SELECT
    std_id,
    quota,
    cycle_used,
    cycle_remaining,
    ROUND((cycle_used / quota) * 100, 1) AS burn_percentage,
    CASE
        WHEN cycle_remaining <= 0 THEN 'Quota Exhausted'
        WHEN cycle_remaining <= 5 THEN 'Critical (Low Quota)'
        WHEN (cycle_used / quota) * 100 > 70 THEN 'Accelerated Burn'
        ELSE 'Optimal Pace'
    END AS pace_status
FROM students;
```

5.3 View: vw_student_active_cycle (In-Progress Cycle Window)

```
CREATE OR REPLACE VIEW vw_student_active_cycle AS
SELECT
    t.std_id,
    t.Transaction_ID,
    t.Status,
    t.Total_Weight,
    t.Drop_Point_ID,
    dp.Location AS drop_point_location,
    t.Drop_Off_Date,
    t.Expected_Completion_Date
FROM (
    SELECT
        Transaction_ID,
        std_id,
        Drop_Point_ID,
        Drop_Off_Date,
        Expected_Completion_Date,
        Total_Weight,
        Cycles_Deducted,
        Status,
        ROW_NUMBER() OVER (
            PARTITION BY std_id
            ORDER BY Drop_Off_Date DESC, Transaction_ID DESC
        ) AS rn
    FROM laundry_transaction
    WHERE Status NOT IN ('Completed', 'Cancelled')
) t
LEFT JOIN drop_point dp ON t.Drop_Point_ID = dp.Drop_Point_ID
WHERE t.rn = 1;
```

5.4 View: vw_student_portal_overview (Composite Dashboard View)

```
CREATE OR REPLACE VIEW vw_student_portal_overview AS
SELECT
    p.std_id,
    p.std_name,
    p.contact,
    p.hostel,
    p.room,
    p.academic_year,
    q.quota,
    q.cycle_used,
    q.cycle_remaining,
    q.burn_percentage,
    q.pace_status,
    a.Transaction_ID AS active_transaction_id,
    a.Status AS active_status,
    a.Total_Weight AS active_total_weight,
    a.Drop_Point_ID AS active_drop_point_id,
    a.drop_point_location AS active_drop_point_location,
    a.Drop_Off_Date AS active_drop_off_date,
    a.Expected_Completion_Date AS active_expected_completion_date
FROM vw_student_profile p
JOIN vw_student_quota_burn q ON p.std_id = q.std_id
LEFT JOIN vw_student_active_cycle a ON p.std_id = a.std_id;
```

6. Schema Teardown & Maintenance DDL

Safe teardown script dropping dependent objects in reverse order of foreign key constraints to prevent orphan references.

```
-- Reverse Dependency Drop Sequence
DROP VIEW IF EXISTS vw_student_portal_overview;
DROP VIEW IF EXISTS vw_student_active_cycle;
DROP VIEW IF EXISTS vw_student_quota_burn;
DROP VIEW IF EXISTS vw_student_profile;

DROP TABLE IF EXISTS transaction_item;
DROP TABLE IF EXISTS laundry_transaction;
DROP TABLE IF EXISTS laundry_item;
DROP TABLE IF EXISTS drop_point;
DROP TABLE IF EXISTS vendor;
DROP TABLE IF EXISTS students;
DROP TABLE IF EXISTS admin_users;
```